# AI & Machine Learning - Quiz Q&A Log

*My Questions, My Answers (in my own words)*

## Topic 1: What is Intelligence?

**Q1. Cousin memorized history = most intelligent?**

My Answer: Nope, intelligence means to take decision based on learning. Memorizing will not count.

**Q2. Cat touches hot stove once, never again?**

My Answer: Yes she is showing intelligence because she knows hot stove will burn her paws.

**Q3. Best example: Calculator / Child avoiding bully / Parrot?**

My Answer: Point B can be, But point C must be.

*Correct: (B). Parrot repeats without understanding.*

## Topic 2: What is AI?

**Q1. "AI means robots that think like humans." Correct?**

My Answer: Correct. If machine/computer thinks like human = AI.

*Correct: "Thinks like humans" right but "robots" wrong. Most AI is software.*

**Q2. Instagram suggests reels. AI?**

My Answer: Learning user past actions, recommends contents.

**Q3. Calculator / Spam filter / Light switch?**

My Answer: Only spam filter. 2+2 is basic rule. Light switch on/off.

## Topic 3: Types of AI

**Q1. ChatGPT - Narrow, General, or Super?**

My Answer: General AI. It does most work by its own.

*Correct: Very advanced Narrow AI.*

**Q2. Alexa does many things = General AI?**

My Answer: Nope, its programmed stuff.

**Q3. Example of Narrow AI?**

My Answer: Recommendation feature in websites. AI note picker in meetings.

## Checkpoint Review #1 (Topics 1-3) - Score: 2/5

**CR1. 4 key parts of intelligence?**

My Answer: Learning from past, make decision, needs data.

*Correct: Learn, Understand, Decide, Adapt.*

**CR2. Difference AI and Robot?**

My Answer: If programmed is intelligent = AI, if repetition = robot.

*Correct: AI = brain (software), Robot = body (hardware).*

**CR3. Self-driving car that can't converse?**

My Answer: Low level Narrow because designed for specific purpose.

**CR4. "General AI exists - ChatGPT proves it."**

My Answer: Yes.

*Correct: False. ChatGPT = Narrow AI.*

**CR5. Google Translate - AI?**

My Answer: Not AI, its a translator.

*Correct: IS Narrow AI. Learns from millions of documents.*

## Topic 4: Traditional vs ML

My Understanding: If I write rules = AI. If data + labels = ML.

*Tutor: Both are AI. Rule-based = AI not ML. Data-trained = AI + ML.*

**Q1. Spam filter - programmer or computer learned?**

My Answer: Computer learns from labelled emails.

**Q2. Face recognition - traditional or ML?**

My Answer: ML, feed many labelled images, AI figures it out.

## Topic 5: AI Hierarchy

*Clarifications: Why DL different from ML? How computer distinguishes faces? What are knobs?*

**Q1. Every ML = AI? Every AI = ML?**

My Answer: Every ML is AI, but every AI is not ML.

**Q2. Spam from 50K emails, simple formula. ML or DL?**

My Answer: Simple ML.

**Q3. Why "deep" learning?**

My Answer: Uses several layers inside, focuses on each detail.

## Topic 6: Supervised Learning

**Q1. Features and labels?**

My Answer: Features = qualities. That something = label.

**Q2. Pass/fail = classification or regression?**

My Answer: Classification. If "how much marks" then regression.

**Q3. Predict exact marks (0-100)?**

My Answer: Regression.

## Topic 7: Unsupervised Learning

**Q1. Supervised vs unsupervised?**

My Answer: Supervised = data + output labelled. Unsupervised = only data.

**Q2. Store groups customers?**

My Answer: Clustering.

**Q3. Card blocked, 10 countries in 1 hour?**

My Answer: Clustering?

*Correct: Anomaly Detection.*

## Topic 8: Reinforcement Learning

*5 clarification questions about RL environment, rewards vs rules, RL vs supervised, exam bot, self-driving car.*

**Q1. YouTube RL: Agent/Environment/Reward?**

My Answer: Agent=algo, Environment=YouTube, Reward=user feedback.

**Q2. Why can't traditional programming teach chess?**

My Answer: Can't make rules to such depth, single move leads to many openings.

**Q3. Kid making chai, mom smiles/frowns?**

My Answer: Reinforcement - based on mother actions, good tea = reward.

## Checkpoint Review #2 (Topics 4-8) - Score: 4/5

**CR1. Traditional - who writes rules?**

My Answer: Human writes rules.

**CR2. Many layers for photo = ML or DL?**

My Answer: Deep Learning, involves layers.

**CR3. Diabetes Yes/No = Classification or Regression?**

My Answer: Regression.

*Correct: Classification. Yes/No = WORD.*

**CR4. Card blocked, 5 countries?**

My Answer: Anomaly detection.

**CR5. In RL, does model know HOW?**

My Answer: Model figures out on its own.

## Topic 9: Neural Networks

**Q1. Which layer receives raw data?**

My Answer: The input layer.

**Q2. 50K images processed once = how many epochs?**

My Answer: Only one epoch.

**Q3. Activation function does what?**

My Answer: Decides if output ok to pass further.

*Tutor: Works at EVERY neuron in EVERY layer, not just final.*

## Topic 10: Math Foundations

*7 clarification questions about why math, vectors, dot product, internal process, model correctness, testing, unsupervised testing.*

**Q1. [3, 1200, 2, 10] = ?**

My Answer: Its called the vector.

**Q2. Gradient descent - top or bottom?**

My Answer: Bottom, narrowing down error = more accurate.

**Q3. Learning rate too big?**

My Answer: Effects outcomes both ways, too high or too low.

*Tutor: Model overshoots - zigzags forever.*

**Q4. 99% training, 30% test?**

My Answer: Wrong data / garbage data.

*Correct: Overfitting. Memorized. Bad data = low on BOTH.*

## Topic 11: How Math Connects to ML

*Q: "Too math, got lost." Tutor re-explained as real estate agent story.*

**Q1. Model learned older houses cheaper. Human taught or data?**

My Answer: Learned from data where it noticed old house sold cheaper.

**Q2. Trained on 5 houses. Predict well?**

My Answer: Not at all, prediction always leads to destruction.

*Tutor: Correct! Too few patterns = weak intuition.*

**Q3. 99% training, 30% test?**

My Answer: Issue with data or threshold.

*Correct: Overfitting! Bad data = low on BOTH. Overfitting = high train + low test.*

## Topic 12: Problem Understanding & Model Selection

*My Request: "I need to understand how to pick the right model, which techniques/tricks lead to best selection."*

### Quiz (Pending)

**Q1. Predict spam/not spam, 5000 labeled emails. ML type? Model?**

**Q2. Camera footage, spot unusual, NO labels. ML type? Model?**

**Q3. Why YOLO for phone detection instead of CNN from scratch?**

**Q4. Random Forest = 82%. Immediately switch to Deep Learning?**